

Embedded tense and the problem of acquisition*

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1. Introduction

It may be strange to begin a paper with a list of what it does not provide. Yet we have, depending how one views it, no positive results to report. We have found no new pattern of acquisition, and we offer no new account of any particular tense-related phenomenon in any language. What we hope to do instead is pose questions that, to our knowledge, have not received attention in the literature. The broad form of the question is all too familiar: how do speakers of a language — say, English — come to possess the grammatical knowledge that they demonstrate? The questions that guide this paper are narrow instances of this broad one, narrowed in three ways. First, they concern three particular languages: Japanese, English, and Russian. Second, they concern tenses specifically, and more specifically still, constructions containing *embedded* tenses. Third, they concern the *semantics* of tense. The learning challenge of interest to us is not merely that the learner must come to know, of certain structures, that they are grammatical or ungrammatical; the learner must come to know what meanings the relevant structures can and cannot have. For instance, one acquiring English must learn that certain instances of embedded past tense permit simultaneous readings, while one acquiring Japanese must learn that they do not. The evidence bearing on such knowledge must consist of a kind of pairing between sentences and the situations they are used to describe, and it is from such pairings (in tandem with their existing grammatical beliefs) that a learner must operate. Before we proceed, we must make it clear that our intended contribution is not specific to the semantics of tense. The issues that we explore are general to the acquisition of any piece of semantic knowledge. We choose embedded tense as a case study.

Any answer to a question about acquisition must begin with explicit assumptions about the learner. The learning framework we describe operates *locally* (both with respect to individual utterances and individual hypotheses) in a sense that we describe and defend

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below (Section 3); with supporting evidence drawn in part from the experimental literature on children’s word learning. A further assumption we highlight concerns where learning begins. Drawing on evidence from Hollebrandse (2000) and Demirdache and Lungu (2011) we assume that children carry an initial belief that the past tense is indexical while the present tense is vacuous — a state which matches none of our three target grammars. The challenge, then, is to arrive at the (adult) grammars of Japanese, English, and Russian from this starting point using our local learning procedure.

Our finding, such as it is, is that the challenge can be met: for each of our languages there is a path from the initial state to the target grammar, provided that certain hypotheses are available to the learner and that certain critical constructions are sufficiently attested in the input. Identifying those critical inputs is, we think, one of the primary contributions of the paper. This is because the inputs that the logic of our account singles out are not necessarily the ones that inspection of the adult grammars would intuitively suggest, and perhaps only come into focus when semantics and acquisition are considered together.

The paper proceeds as follows. We first review the crosslinguistic facts about embedded tense in Japanese, English, and Russian that constitute our targets of acquisition. We then lay out the learning framework and defend its assumptions. Finally, we trace a path from the shared initial state to each of the three adult grammars, and close by discussing some predictions, and vulnerabilities, of the account.

2. Embedded tenses in Japanese, English, Russian

The empirical claims we review in this section come mostly from Ogiwara and Sharvit 2012, O&S hereafter. We must make it clear that our coverage of the data will be neither exhaustive nor balanced. We limit our attention to specific findings only so we can describe what some of the relevant crosslinguistic differences mean for acquisition.

We start with (1). (1a-b) are reasonable first hypotheses about the semantics of the past tense and the present tense. The denotations operate on a time argument t and a propositional argument p . In the case of PAST, t and p are mapped to True iff p holds at a time that precedes t ; in the case of PRES, p must hold at t .

- (1) a. $\llbracket \text{PAST} \rrbracket = [\lambda t . \lambda p . \exists t' (t' < t \ \& \ p(t') = 1)]$
 b. $\llbracket \text{PRES} \rrbracket = [\lambda t . \lambda p . p(t)]$

With these entries, and additional innocuous assumptions about semantic composition, we expect the past to require back-shifting locally, and the present to introduce no shifting. This means that a sentence where a past-marked clause is embedded under a past-marked attitude verb, like (2), should say (i) that the relevant attitude held at a time before the time of speech, and (ii) that the content of the attitude is that some state of affairs held before the perceived present of the attitude holder.

- (2) Bill believed that it was raining.

From (1b), our semantics of PRES, we predict that a sentence where a present-marked clause is embedded under a past-marked attitude verb, like (3), should say (i) that the relevant attitude held at a time before the time of speech, and (ii) that the content of the attitude is that some state of affairs holds *at* the perceived present of the attitude holder.

- (3) Bill believed that it is raining.

In the rest of the paper we will sometimes use the term ‘before zero’ (in symbols, $<\emptyset$) to refer to the local back-shifting semantics of PAST in (1a), and the term ‘zero’ ($\emptyset$) to refer to the non-shifting semantics of PRES in (1b). Soon we will see that the predictions from (1) are met in Japanese and Russian, but not in English.

Another important part of our dataset features the future marker. We will assume that the future shifts evaluation time in the same way as the past, but in the opposite direction:

- (4) $\llbracket \text{FUT} \rrbracket = [\lambda t . \lambda p . \exists t' (t < t' \ \& \ p(t') = 1)]$

(4) leads to predictions that mirror the ones described above. A sentence like (5) should say that the relevant attitude holds at a time after speech time, and that its content is that the relevant state of affairs takes place after the attitude holder’s perceived present.

- (5) Bill will think that it will be raining.

This seems to be correct, but we will not be concerned with examples like this. We will, however, pay attention to sentences like (6), where the future embeds the present.

- (6) Bill will think that it is raining.

Again, the prediction mirrors the case where the present is embedded under the past. Sentences like (6) should say that the content of some future attitude is that the relevant state of affairs takes place at the perceived present of the attitude holder. This time, we find that the prediction is met in all three languages, not just in Japanese and Russian.

In the next three subsections we will say more about the similarities and differences between our three languages, and as we do, we will revisit the semantics in (1) as needed.

2.1 Japanese (and a preview of the paper)

Japanese comes closest to matching the semantics in (1), assuming that t in the entries is local evaluation time. In (7), we show an example of a PAST>PAST embedding. Sentences like this seem to have only the (predicted) back-shifted reading.

- (7) Taroo-wa [Hanako-ga byooki-dat-ta] -to it-ta.
Taro-TOP Hanako-NOM be-sick-PAST that say-PAST
‘Taro said “Hanako was sick”’

(8) is an example of a PAST>PRES embedding. Here, as the translation indicates, we also find confirmation of (1b); the present tense has the $\emptyset$ -interpretation.

- (8) Taroo-wa [Hanako-ga byooki-da] -to it-ta
 Taro-TOP Hanako-NOM be-sick-PRES that say-PAST
 ‘Taro said “Hanako is sick”’

Things become more complicated when we look at embedding across relative clauses (RCs). In these cases, PAST>PAST configurations permit back-shifted readings as well as simultaneous and forward-shifted readings. (We find the same three-way ambiguity in English and Russian also, as we will soon see.)

- (9) 1989-nen ni, Taroo-wa 1970 nen dai ni / sono toki / 1990 nen dai ni
 1989-year in Taro-TOP 1970s in / that time / 1990s in
 aisi-tei-ta zyosei-ni at-ta
 love-PROG-PAST woman-DAT meet-PAST
 ‘In 1989 Taro met a woman who in 70s / then / in 90s loved him’

Here we already have something of a puzzle. Imagine a child growing up in a Japanese-speaking community. What experience tells the child that (9) allows the simultaneous reading? One can only presume that it is hearing enough examples that look like (9), and that either contain temporal modifiers that force the simultaneous reading, as (9) does, or occur in contexts where only the simultaneous reading makes sense. Many details of this answer deserve a closer look of course, but the question that we want to highlight is why the learner does not conclude from (9) that (10) — not (1b) — is the correct semantics of the past.

- (10) $\llbracket \text{PAST} \rrbracket = [\lambda t . \lambda p . \exists t' (t' < u \ \& \ p(t') = 1)]$ (where u = utterance time)

(10) is an indexical semantics of the past tense ($\text{PAST}_{<u}$); it says that the shifting introduced by the past tense is relative to the time of utterance u , not local evaluation time. On such a semantics, (9) would correctly be predicted to permit a simultaneous reading — in fact, it would be predicted (again correctly) to also permit a back-shifted and a forward-shifted reading. But (10) would also allow (7) to have a simultaneous reading, and the adult grammar of Japanese does not. Note the shape of the problem here: in RCs the two hypotheses generate the same readings, while under attitude verbs the readings licensed by (1a) are a proper subset of those licensed by (10). This is the subset problem (*a la* Berwick 1985). So either children do not entertain (10) at all, or they do but they move away from it later in their linguistic development. Neither of these possibilities is obvious.¹

¹We will come back to (10) when we review Hollebrandse’s (2000) study of the acquisition of SOT. Hollebrandse’s findings suggest that there is an early stage of development where children assign sentences like (7) the same readings as (9): a back-shifted, simultaneous, and forward-shifted reading. This suggests that children do in fact have an indexical semantics of the past. The question is how they move away from it.

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Let us return to the data. When the Japanese present appears inside an RC, it can have the zero reading. For example, (11) can mean that Taro, in his childhood, met a woman who loved traveling at the time when they met. Neither Russian nor English allow this interpretation when the embedded tense is PRES, as we will see shortly.

- (11) Kodomo-no koro, Koji-wa [ryokoo-o aisu-ru zyosei] -ni at-ta
child-GEN time, Koji-TOP traveling-ACC love-PRES woman] -DAT meet-past
'In his childhood Koji met a woman who loved traveling'

So far the data tell us that, in Japanese, PAST and PRES have the $<\emptyset$ and $\emptyset$ semantics. When one occurrence of PAST embeds another across an attitude verb, the semantics forces back-shifting. When the PASTs are separated by an RC boundary, shifting is not required. This is consistent with a $<\emptyset$ semantics of PAST, as long as the temporal perspective in the RC is not assumed to shift obligatorily with the matrix PAST. This freedom from shifting would follow if, say, the noun phrase that contains the RC takes scope above the matrix PAST. In that case, neither tense would be in the scope of the other, and they would each shift backwards from their local evaluation time, i.e. utterance time. With the present we have more or less the same picture. When PRES is embedded under a PAST-marked attitude verb, or under a FUT-marked one (example below), the PRES behaves as if it were vacuous.

- (12) Taroo-wa [Hanako-ga byooki-da] -to i-u (daroo)
Taro-TOP Hanako-NOM be-sick-PRES that say-NONPAST (EVID)
'Taro will say "Hanako is sick"'

The same possibilities are available when PRES is embedded inside an RC, but in that case, PRES can point to utterance time, i.e. to $\emptyset$ with respect to the root-level evaluation time.

- (13) Tyuunen-ni nat-te kara Koji-wa [yat-to tabi-o aisu-ru
middle.age-DAT become-from Koji-TOP [finally traveling-ACC love-PRES
zyosei] -ni au-daroo
woman] -DAT meet-probably
'In middle age, Koji will finally meet a woman who loves traveling'

2.2 English

Embedded tenses behave differently in English. First, English PAST>PAST embeddings allow simultaneous readings both across attitude verbs and across RC boundaries. To our knowledge, the most widely accepted account of this in the field is agreement-based: English is a so-called "SOT" language, that is, a language in which the morphosyntactic realization of a tense can be (in a manner of speaking) the product of agreement with a higher

occurrence of that same tense.² This allows (14), repeated from (2), to have a simultaneous reading (but not a forward-shifted reading), in addition to its back-shifted reading:

- (14) Bill believed that it was raining. (=2)
 ≈ Bill's belief at some earlier time: "it is raining"

Let us make it clear that, when we say that a language is an SOT language, we refer not merely to the fact that the language permits simultaneous readings of, say, PAST>PAST constructions. Rather, we mean that the language permits this reading *because* the embedded tense in this construction is semantically vacuous ($\emptyset$) and is realized morphologically by an agreement mechanism with a higher tense (a "fake" PAST/PRES under a higher PAST/PRES).

The Japanese analog of (14) does not permit the simultaneous reading, so whatever makes the reading available in English — whether it is SOT or something else — must not operate on tenses in Japanese. In the next section we will see that Russian is mixed, something that complicates the typology (as O&S point out), and therefore the learning task.

The present tense in English is also different from Japanese. Neither (15) nor (16), analogs of the Japanese (11), allow the embedded present to take the $\emptyset$ -reading. Here, PRES must be interpreted as the time of utterance. Notice first that (15) is acceptable, and therefore that the embedded PRES can be understood as referring to utterance time:

- (15) I went to school with someone who is doing time in Sing Sing.

Notice also that (16) is odd; it seems to intuitively suggest that Bill's classmate has lice now.

- (16) #Bill went to first grade with someone who has lice.

Although there is nothing wrong with this proposition logically, the mention of first grade seems to favor an interpretation where the lice problem is contemporaneous with the time of being in first grade, i.e. a $\emptyset$ -reading of the embedded present. English does not seem to allow this interpretation, unlike Japanese.

Finally, (17), which is acceptable, requires the content of Bill's report to hold true (according to Bill) at utterance time in addition to Bill's perceived present. This is the "double-access" reading:

- (17) Bill told us that Mary is sick.

The obligatoriness of the inference to utterance time is noticeable in the following contrast:

²Implementations differ in how the agreement is achieved, e.g., via deletion at LF (Ogihara 1996) or feature transmission under binding (Kratzer 1998), etc. We will propose a variant of the latter under which a $\emptyset$ -interpreted tense is a lexical entry in its own right. We will nonetheless continue to use the term "agreement" broadly, for whatever process aligns the time and morphology of the embedded tense with that of a higher tense. For the typological discussion here nothing hinges on this choice. What matters, most centrally, is whether a language makes available a $\emptyset$ reading, and in which environments.

- (18) I don't know how old Mary is, but when I asked Bill last week, he told me that she is in her early thirties.
- (19) I don't know how old Mary is, but when I asked Bill ten years ago, #he told me that she is in her early thirties.

The contrast here is driven by the difference in how much time separates Bill's report (his perceived present) from utterance time: one week in (18); ten years in (19). A person can be (believed to be) in their early thirties for one week, but not for ten years. This is what makes (19) odd. Crucially, the oddness depends on an indexical semantics of the English present, not the $\emptyset$ -semantics that the Japanese data suggest.³ A simple entry for the English present may look something like this:⁴

- (20) $\llbracket \text{PRES} \rrbracket = [\lambda t . \lambda p . t \text{ overlaps with } u \text{ and } p(t) = 1]$

Still, there are cases where the English PRES takes the $\emptyset$ -interpretation. These are sentences where PRES appears embedded under a present-marked future operator, like *will*. (21) and (22) are examples:

- (21) When I ask Bart a question tomorrow, he will tell me that he is not feeling well.
≈ Tomorrow, Bart will say "I am not feeling well"
- (22) When you move to New York, you will have roommates who never pay rent.

On the assumption that *will* is itself a present-marked operator, the $\emptyset$ -reading of the embedded present here can be treated as another instance of SOT-agreement.

Let us summarize: there are at least two important differences between English and Japanese. First, the present tense in English is indexical, but in Japanese it is a $\emptyset$ -tense. Second, English can be described as +SOT, so an embedded instance of PRES (or PAST) in English can have a $\emptyset$ -reading when it appears under another instance of PRES (or PAST). We will bring up a few additional complexities when we return to questions of acquisition.

2.3 Russian

We finally turn to Russian. As we said, Russian appears to be like Japanese in some respects, and like English in others. It is like Japanese in requiring back-shifting when a PAST-marked attitude verb embeds a PAST-headed clause. (23) is an example.

- (23) Ivan skazal čto Maša bolela.
Ivan said that Masha ailed
'Ivan said "Masha was sick"'

³Japanese and Russian analogs to (18-19) are equally acceptable. We omit the examples to save space.

⁴There is more to double-access than what we reviewed here. We refer readers to Abusch 1997.

It is also like Japanese in having a $\emptyset$ -present; (24) is like the Japanese (8):

- (24) Ivan skazal čto Maša boleet.
 Ivan said that Masha ails
 ‘Ivan said “Masha is sick”’

These datapoints suggest that Russian is a –SOT language, and that it has a $\emptyset$ -present. However, RC embeddings complicate the picture. First, PRES does not have the $\emptyset$ -reading in RCs; in (25), the embedded PRES must be about utterance time. In this, Russian is like English and unlike Japanese.

- (25) Ivan uvidel devočku kotoraja sidit na skamejke
 Ivan saw girl which sits on bench
 ‘Ivan saw a/the girl who is sitting on the bench’

Second, PRES can have the $\emptyset$ -reading under a (PRES-marked) future operator, as in (26):

- (26) (Čerez desjat’ let) u Ivana budet rebėnok kotoryj mnogo est
 In ten years to Ivan will.be child which much eats
 ‘In 10 years, Ivan will have a child who eats a lot.’

Since the eating-a-lot is in the future — contemporaneous with Ivan’s having the child — the embedded PRES must be interpreted as $\emptyset$. But from (25) we know that PRES does not always accept the $\emptyset$ -reading (unlike in Japanese). What then explains the difference between (25) and (26)? We take from O&S the proposal that $\pm$ SOT is not a language-wide parameter, or at the very least, that it is subject to conditioning. Our proposal, which we spell out below, will be that the availability of $\emptyset$ -reading is stated in the lexicon. How learners could reach target grammars of this kind is the question that occupies the rest of the paper; we turn to it after summarizing the typology.

2.4 Summary in terms of parameters

The following table summarizes the facts we reviewed:

	Japanese	English	Russian
SOT	–	+	– (PAST)
			+ (PRES)
PRES	$\emptyset$	<i>u</i>	$\emptyset$ (under AVs)
			<i>u</i> (elsewhere)
PAST	< $\emptyset$	< $\emptyset$	< $\emptyset$

Two notes before we turn to our learning model. First, the bottom row of the table, which records no crosslinguistic variation, is included because we will later return to what

we called the indexical past ($<u$ —recall (10)). Second, although our review has been faithful to the literature in treating SOT as a syntactic mechanism, we will depart from this characterization in our model. Our implementation of SOT will be lexical: a tense morpheme may be associated with more than one entry, each (potentially) restricted to particular licensing environments. So, in English, PAST is ambiguous between the back-shifting entry in (1a) and a $\emptyset$ entry. The second of these is only licensed under another PAST. A +SOT language, on this view, is simply one whose lexicon contains a $\emptyset$ entry for the relevant tense; a –SOT language is one whose lexicon does not. During acquisition, $\emptyset$ entries may be added to the lexicon (where appropriate) alongside the shifting ones, rather than instead of them. This will enable us to capture the optionality of SOT, and make the semantics of tense learnable within a plausible framework of acquisition.

3. The learning framework: Grammatical Pursuit

We outline a framework for the acquisition of the semantics of tense⁵ which we call *Grammatical Pursuit*; building on Stevens et al.’s (2017) Pursuit model of word learning. Learning is local to individual lexical entries, and incremental across individual utterances.

Throughout, we will try to distinguish our methodological assumptions—adopted for tractability (though doubtless true in some form) and relaxable without changing the character of the account—from our foundational assumptions, which encode substantive claims about the learner and which we defend by appeal to what is known about both acquisition and memory. We aim to draw out a connection between the acquisition of embedded tense and the large base of experimental and computational work on how children attempt to resolve referential ambiguity and learn the meanings of individual words.

We model learning as a sequence of local updates to the grammar. At any given time the learner has knowledge of a candidate grammar g , which contains a set of lexical entries $\{r_1, \dots, r_n\}$. The entries of interest to us associate a tense form, in a conditioning environment, with an interpretation. The environment may be trivial (“anywhere”) or restricted (“under PAST”, “under an attitude verb”). Each entry additionally carries a numeric score $c(r)$, which one can think of as the learner’s confidence that the entry is correct. Entries enter the grammar as members of a *competition*: a set of distinct entries, of which only one may ultimately be deemed satisfactory.⁶ How the grammar deploys these scored entries in the face of an utterance, and how the scores change in response, we describe presently.

At each step the learner is presented with an input and evaluates it against the grammar they are currently entertaining. The object evaluated at each step is a triple, $\langle \sigma, m, r_s \rangle$, consisting of (i) a parsed utterance σ , (ii) the meaning m detected on that occasion, and (iii) an entry r_s under evaluation whose conditioning environment σ instantiates.

⁵Though the procedure is sufficiently general that it may, potentially, be applied to the learning of any comparable form-meaning mapping.

⁶As we develop further below, there are situations in which the learner may be forced to conclude that the grammar contains lexical ambiguity in the semantics of tense. i.e., two entries the learner had been treating as rival hypotheses may, in fact, both be required for capturing the range of linguistic inputs (the morphologically realized tense is lexically ambiguous). At such a point the rules cease to compete and instead are taken to manifest separate homophonous tenses, with their own (possibly trivial) competition.

Parsing σ identifies the embedded tense together with the configuration it appears in (e.g., under an attitude verb, or across an RC boundary), and this pairing selects an entry “slot” s . For each slot, the learner’s grammar comprises a set of competing candidate entries R_s , from which the top-ranked entry ($r_s = \arg \max_{r \in R_s} c(r)$) is deterministically applied. It is this front-runner that is evaluated in the potentially updating triple. The grammar that governs interpretation at any moment is thus the collection of front-runners across slots, drawn from the fuller slate of scored competitors under active consideration.

3.1 Evaluation and updating

The evaluation of an input triple has one of two immediate results. The input is *consistent* with the top-entry r_s when applying r_s to σ —in the context of the rest of the current syntactic and semantic knowledge of the learner—licenses the detected meaning m , and *inconsistent* otherwise. Consistency rewards r_s (increasing the score attached to that entry $c(r_s)$). Inconsistency penalizes r_s (decreasing its score) and in addition selects (at random) an alternative entry r' from the hypothesis space to reward—one consistent with the parsed utterance σ and the meaning m —adding r' to R_s if necessary, and increasing its score.⁷

In the simplest case, the front-runner is consistent with (nearly) all the inputs to its slot and so the learner treats this as a stable state. Crucially, a single inconsistent input does not necessarily lead the learner to abandon r_s ⁸, and thus may not meaningfully update the grammar g : the applied grammar is unchanged so long as the arg max at slot s holds.

Thus an entry may remain the learner’s favored hypothesis for its slot despite the occasional counterexample. As inconsistent inputs recur, however, the score $c(r_s)$ erodes, while a competing entry $r' \in R_s$ —one that has been rewarded on the inputs that r_s fails—gains. Eventually the competitor $c(r')$ overtakes $c(r_s)$, the front-runner for the slot changes, and with it the grammar g that is next applied. This reward-and-penalty dynamic is adopted from Stevens et al.’s Pursuit model; a form of greedy probabilistic reinforcement learning.

Finally, it may be that no single hypothesized denotation is consistent with all of the attested input. This is true of PAST in English for instance, which as introduced above is generally $<\emptyset$, except for instances subject to SOT where it may be interpreted as $\emptyset$. Neither meaning on its own suffices to capture the data any more than a single meaning of a homophonous wordform could apply to all necessary contexts. In such circumstances the two update operations produce a particular dynamic: whichever entry currently leads the competition is penalized on precisely the inputs that reward its rival, so that no lead is stable, and the front-runner position alternates indefinitely. This persistent oscillation between

⁷Note that more than one alternative may be consistent with a given failure, so the learner has no way of uniquely identifying, from any single occasion, what the correct repair is. We further note, but leave to future work, the question of whether there are any governing principles controlling how or which alternative entries are selected or generated from the hypothesis space when needed. Such principles may explain behavior that could be described as “aggressive” vs. “conservative” generalization on the part of the learner.

⁸The maximal size of R_s , the number of competing entries per slot, may be considered. We posit no particular restriction going forward, but its lower limit is instructive. In the extreme case where $|R_s| = 1$ is enforced, the learner keeps no scored competitors at all. This holds a single entry per slot and, on an inconsistent input, must replace it outright rather than let a rival accrue confidence. This is a semantic counterpart of parameter *triggering* in the sense of Gibson and Wexler (1994), and, as an idealization of learning, an overly brittle one.

unstable states may itself be informative, evidence for a learner that the mutual exclusivity between this pair of hypotheses must be relaxed. The erstwhile rivals are re-analyzed as two tenses sharing a morphological realization, each heading its own slot, and each now free to converge within its own (possibly restricted) environment. We acknowledge that this description leaves the trigger imprecise — we have not specified how much oscillation, over what span of input, licenses such re-analyses — yet for practical reasons we proceed with the informal characterization for the time being.

3.2 Practical assumptions

Each component of the input triple encodes an assumption, and we take each of them in turn. The first, receiving the parsed utterance σ , is likely the less controversial. σ is the one component recoverable from the signal itself, requiring only that the learner parse well enough to detect an embedded tense, and under what matrix configuration the embedded tense resides. Taking this parse as given is not wholly innocent, since it presupposes progress in acquiring some relevant syntactic/semantic knowledge, but that problem is separable from, and logically prior to, the semantic one at issue here. We set it aside.

The second assumption is that, on a given occasion, the learner’s detected meaning m is in fact the speaker’s intended meaning. Here this is primarily the temporal configuration of the tense-bearing predicates (e.g., back-shifted, or simultaneous) the utterance was used to convey. This is plainly an idealization, but two observations blunt its force. First, the assumption is needed regardless of framework. Any theory of semantic acquisition must pair forms with meanings, and none can proceed if meaning is wholly unavailable. Second, the assumption need not hold universally, but only on some reasonable proportion of occasions. When the learner misidentifies or fails to recover m , the effect is noise (it is exactly the job of the learner to generalize in spite of this) and thus a smaller pool of usable evidence.⁹ The incremental update to entry scores is exactly what provides tolerance to noise.

3.3 Foundational assumptions: Local learning two ways

We now want to defend our decision to take learning to be local. By “local” we mean two things, both true of the procedure outlined above. First, the procedure is *incremental* across utterances: its individual updates to the grammar are responses to the given input alone — an instantiation of $\langle \sigma, m, r \rangle$. The learner does not retain past inputs for subsequent reinterpretation. Second, evaluation is “greedy” (limited) to a single hypothesis r_s : at each step only the front-running entry for the relevant slot is evaluated — and, on failure, a single competitor rewarded — rather than the full candidate set in parallel.

A learner that does not adhere to incrementality (i.e., a batch or global learner) could evaluate hypotheses against an accumulated record of experience, returning to utterances encountered long before and re-scoring them under hypotheses entertained only later. This is the regime that idealized cross-situational or ideal-observer models presuppose (Frank

⁹The same logical problem arises in word learning. There, too, a majority of contexts and utterances, even for common concrete nouns, are referentially underdetermined (Medina et al. 2011:*inter alia*) and learners must exploit the relatively rare, particularly informative contexts.

et al. 2009)¹⁰ to draw batch, corpus-level inferences (Perfors et al. 2011), and it makes an exceedingly heavy demand on memory that we must unpack. Global learning requires that past inputs be retained as individuated form-meaning episodes, each preserved intact enough to be revisited and re-scored. There is little reason to think that memory supplies anything of the sort. While human memory can, of course, be shockingly durable (Bahrick et al. 1975), this depends on the format of information involved. Memory for acoustic-phonetic information, for instance, does not persist in an actionable form after an interval of only a single second (Caplan et al. 2021), while memory for phonological and lexical categories may be maintained for orders of magnitude longer. Classic studies of sentence memory (e.g., Bransford and Franks 1972) show that within moments we retain the integrated event semantics of what we heard while losing its morphosyntactic form, confidently “recognizing” never-before-heard sentences so long as they express the right general content. What is encoded and persists, in other words, is the general meaning, but we are surprisingly poor at retaining the exact form of utterances. But acquiring the semantics of tense requires registering which form — which marker, in which configuration — carried a given temporal relation. This is not to deny that learners accumulate information across experience; they plainly do, and learning cannot be a succession of isolated eureka moments. Our assumption around incrementality concerns only the representations over which evidence accumulation may operate. Under Grammatical Pursuit the state of a learner’s knowledge is carried by the scores attached to competing entries (memory for hypotheses and for their relative support), without direct access to the distribution of utterances that shaped them.

Our second commitment to locality, that evaluation is limited to a single hypothesized entry at a time (in contrast with parallel evaluation of the set of potential entries), finds support from the experimental literature on cross-situational word learning. In Trueswell et al. (2013), for instance, participants encountered novel words across a sequence of referentially ambiguous scenes. No single scene fixed a to-be-learned word’s meaning, but one referent recurred reliably, and participants registered a provisional guess on each round. A learner evaluating hypotheses in parallel — crediting every co-present referent on every trial — would accumulate support for the recurring referent even when guessing wrong (like a tense entry consistent with the input even if it is not currently the top-ranked choice). Participants do not behave this way. Rather, the reward-and-penalty dynamic applies to only the top-ranked hypothesis (inferred via the provisional guess). The critical diagnostic lies in what happens after a wrong guess. Because the target referent recurs on the following scene, a learner who had retained partial evidence for non-top alternatives should reliably select them; instead, performance sits *exactly* at chance. The case for “greedy” evaluation of a single meaning is strengthened with scale: the lab task spans just a few minutes (with alternatives visually present on screen) but even proof-of-concept evidence for parallel evaluation is lacking. The semantics of tense is acquired across a timescale measured in years. Learners could evaluate *anything* (in the hypothesis space) that the data support, but they can’t evaluate *everything*.

¹⁰Cf. Caplan (2025) for an incremental derivation of some ideal-observer effects in word learning.

The two locality commitments together make the learning process finite-state¹¹ (obeying the Markov property), with updates dependent on two objects: the current grammar and the present input. A consequence we want to draw out is this: A given $\langle \sigma, m \rangle$ input does not, on its own, trigger any particular grammar. Whether it drives a transition at all — and if it does, to what — depends on the state the learner is already in. “Triggering,” even in an informal sense, is a property neither of a sentence, nor of a sentence-situation pairing, but of the full triple (sentence-situation-grammar). The course of acquisition is therefore fixed jointly by where the learner begins and by receiving the relevant input data that may carry it, state by state, to the target grammar. In the next section we first review evidence bearing on where learners of tense in fact begin — the initial hypotheses children entertain, insofar as the experimental record reveals them — and then trace, within Grammatical Pursuit, a path from that starting point to our three target adult grammars.

4. Getting to Japanese, English, and Russian

4.1 What is children’s “initial” state?

We take the learner’s initial state to be one in which the present has a $\emptyset$ -semantics ($\text{PRES}_{\emptyset}$) and the past an indexical semantics ($\text{PAST}_{<u}$). There is limited empirical work on the acquisition of the semantics of tense, but that which can be known from the experiments in the existing literature we must take seriously. As no single study has examined responses to each relevant sentence-situation pair across the languages in question, a picture of the initial state must be inferred from their consistently shared behavior. Thankfully it is consistent: children well into development — the 5-to-6-year-olds who constitute the representative learners below — reliably construe embedded tenses differently from adults, accepting them in situations that adult speakers of the same language reject, and in particular ways. Crucially, where multiple languages have been tested, we find convergent behavior among learners, with child judgments aligned with each other *regardless* of the target adult grammar. This initial behavior therefore appears not to be a reflex of the input, which necessarily differs across languages that differ in the semantics of tense and thus their usage. And yet learners of each of these languages evidently do arrive at their respective targets — which is to say that a learner beginning from an assumed state in which PAST is indexical and PRES is $\emptyset$ must be able to reach the adult grammars of Japanese, English, and Russian alike.

We take learners’ initial $\text{PAST}_{<u}$ -state on the basis of Hollebrandse (2000), who tested child learners of English, Japanese, and Dutch on past-over-past sentences like (27),

(27) Cookie Monster said that his animals were in the barn.

Hollebrandse varied the described context so that the embedded event was either anterior to the matrix event, simultaneous with it, posterior to it (but anterior to utterance time), or posterior to both it and utterance time. Children up to at least age six years old, in all

¹¹The idea is that grammars are states (collections of top-ranking entries) and transitions are the input-driven changes that potentially replace one top-ranking entry with another.

three groups, accepted the first three readings while rejecting the fourth.¹² This profile imposes exactly one boundary, namely that the embedded event must precede utterance time irrespective of the matrix time. This is exactly what $PAST_{<u}$ provides.

As for the present, Demirdache and Lungu (2011) tested 5-to-7-year-old learners of French on past-over-present sentences like (28), in contexts where the embedded state had ended by utterance time (a bird receives bread while in a car, then clearly flies away):

- (28) Anne a donné un morceau de pain à l’oiseau qui *est* dans la voiture
 “Ann gave a piece of bread to the bird that *is* in the car.”

The adult French present is indexical, like in English, and cannot denote an interval lying wholly in the past. Nevertheless children overwhelmingly accepted sentences like these on a simultaneous reading, in RCs and complement clauses alike, suggesting a $\emptyset$ -present.

The initial state ($PAST_{<u}$, $PRES_{\emptyset}$) matches none of our three targets, albeit divergent to varying degrees. The paths from this shared origin to the target grammars accordingly differ in length and in the critical inputs they require. We walk through each in turn.

4.2 Japanese

The simplest part of our three-language case study is Japanese. Recall that in (adult) Japanese, the present tense appears to have a $\emptyset$ -semantics, and the past tense a $<\emptyset$ -semantics. If we take the findings reviewed above to show a $PRES_{\emptyset}$ - $PAST_{<u}$ initial state, then the most pressing question is whether there is positive evidence that is inconsistent with $PAST_{<u}$, and that therefore pushes the learner towards $PAST_{<\emptyset}$. It is obvious that the answer does not lie in monoclausal sentences, since the meanings of these are consistent with both $PAST_{<u}$ and $PAST_{<\emptyset}$. The only embedded instances of $PAST$ that contradict $PAST_{<u}$ are those that are (understood to be) about times that do not precede utterance time. These are cases where $PAST$ appears under a forward-shifting operator, as in (29-30):

- (29) Bill will most likely tell us, yet again, that there was traffic on his way.
 (30) After her field trip next month, she will show us the data that she collected.

Japanese constructions like these permit the same past-from-future readings. The following were given to us by Yasutada Sudo:

- (31) (ashita) Taroo-wa [(*pro*) byooki-dat-ta] -furi-o su-ru (daroo)
 (tomorrow) Taro-TOP (*pro*) be-sick-PAST -pretense do-NONPAST (EVID)
 ‘Taro will pretend that he was sick’

¹²Hollebrandse (2000): p. 127 (Dutch), p. 144 (English), p. 155 (Japanese). Evidence of rejecting the forward-shifted reading emerged only in some 7-year-olds. It should be noted that the youngest (3-4 y/o) children tested additionally accepted a future-shifted, *after* utterance time, reading — consistent with not assigning any consistent meaning to past-tense morphology — though this was attenuated by age 5-6 y/o.

- (32) ryokoo-no atode, kare-wa [*pro t* atsume-ta] deeta-o misete-kureru daroo
trip-GEN after, he-TOP [*pro t* gather-PAST] data-ACC show-BENF EVID
‘After his next trip he will show us the data he collected’

So if indeed children initially hypothesize $PAST_{<u}$ (as Hollebrandse’s findings suggest), then examples like these are critical: they lead Japanese learners away from $PAST_{<u}$ and therefore away from allowing simultaneous readings of $PAST > PAST$ embeddings.

This, for our purposes, is the only change involved in moving from the “initial” state to the adult Japanese grammar. We must add that, from the existing experimental data, we do not know at what age learners of Japanese typically become adult-like in their command of embedded tense; we only know that the age is above 5-6 y/o. Relatedly, we do not know how frequently examples like (32) occur in the input data. If it turns out that such future-past sentences are rare, it is not clear how they can form the consistent basis for learning that $PAST$ is not indexical. Conversely, if they are abundant, it is not clear why it takes children so long before they move away from an indexical $PAST$ grammar.

4.3 English

Turning to English, the target grammar is one where $PAST$ is non-indexical — at least sometimes — $PRES$ is indexical, and both are subject to SOT. We saw in the previous section how moving from $PAST_{<u}$ to $PAST_{<\emptyset}$ is possible — the key example is (32). The questions that will occupy us here are, (a) how it is possible to move from $PRES_{\emptyset}$ to $PRES_u$, and (b) how it is possible to learn SOT. Neither is simple, as we will try to show.

What English sentences are compatible only with $PRES_u$? Not monoclausal $PRES$ -headed sentences, of course, since these are compatible with both $PRES_{\emptyset}$ and $PRES_u$. But also, we do not see how sentences where the present is embedded under an attitude verb could be any help. These sentences (in English) either have a $\emptyset$ -present, e.g. if the attitude verb is present/future-marked, or have the double-access reading, if the attitude verb is $PAST$ -marked. But the double-access reading *entails* the reading that would result from a $\emptyset$ -interpretation of the embedded present (recall our discussion of double-access in Section 2.2). So, any situation that warrants the use of such a sentence is one where the embedded present can be understood as $\emptyset$.¹³ That leaves only cases in which the present is embedded under the past across an RC boundary, as in (33), repeated from (15):

- (33) I went to school with someone who is doing time in Sing Sing.

So now we have two critical inputs for English: one for $PAST_{<\emptyset}$, and one for $PRES_u$.

¹³One might think that the contexts prompting double-access sentences could themselves signal indexicality. However, it is relevant that in Russian, even in situations where the speaker wants to draw attention to a belief or report about utterance time, the embedded present is used. For example, if a Russian speaker wants to say that Bill said that Mary is in town, where context makes it clear that the subject matter is whether (according to Bill), Mary is in town *at utterance time*, the speaker would use a $PAST > PRES$ embedding. Since this is true of Russian, and since Russian learners successfully acquire the detail that $PRES_{\emptyset}$ is licensed under attitude verbs, it follows that these sentence-context pairings do not drive learners to $PRES_u$.

What about SOT? Recall that SOT is optional, and that optionality in our paper is modeled as lexical ambiguity: a “deleted” instance of PAST is $PAST_{\emptyset}$, which is represented in the lexicon as having the meaning $\emptyset$ and as being licensed only under higher occurrences of PAST. What input motivates positing a slot for $PAST_{\emptyset}$? A tempting answer is $PAST > PAST$ embeddings that (are understood to) have the simultaneous reading. (34) is an example:¹⁴

(34) I was quiet because I thought you were asleep.

Obviously, $PAST_{\emptyset}$ must not replace $PAST_{<u}$; the two entries must exist simultaneously. And because (we assume) there are enough examples that motivate $PAST_{<u}$ and $PAST_{\emptyset}$, the learner (as we model them) posits both. We write this ambiguity as $PAST_{<u/\emptyset}$.¹⁵

Now, (34) is also compatible with $PAST_{<u}$. So a learner who posits $PAST_{<u/\emptyset}$ in response to (34) could have just as well posited $PAST_{<u/<u}$ instead. In fact, given that $PAST_{<u}$ is (at least by assumption) part of the initial state, there may be good reason to think that learners will resort to it, rather than $PAST_{\emptyset}$, when they encounter sentences like (34). If this is right, then something else must drive learners to add $PAST_{\emptyset}$ to their lexicon.

Here we find another puzzle. The puzzle is not so much whether learners encounter examples that require $PAST_{\emptyset}$ — we are willing to grant that they do. The puzzle comes from how $PAST_{\emptyset}$ and $PAST_{<u}$ interact. First, let us make it clear that both entries must exist in adult English. The following are examples that can only be analyzed with $PAST_{\emptyset}$:

- (35) a. Bill promised to brush his teeth before he went to bed.
 b. Bill said last week that in ten days he would tell his mother that they were having their last meal together.
 c. Mary said that, whatever happened, she would stand by her friends.

And (36), modeled after Stowell 2014, can only be analyzed with $PAST_{<u}$.

- (36) My grandfather, who died before Watergate, thought that Nixon was a good president. These days, hardly anyone does.

Target reading: My grandfather’s opinion: “Nixon is a good president.”

Current opinion: “Nixon *was* not a good president.”

The second part of (36) has an elided VP. Assuming that semantic identity is a condition on VP ellipsis, it follows that the two VPs must contain (semantically-)identical complement clauses. If the complement clauses contain a zero tense, then the two parts of (36) should have the simultaneous reading: my grandfather thought that Nixon was a good president at his perceived present, and these days nobody thinks that Nixon (now) is a good president. This is not the target reading. Alternatively, if the complement clauses contain $PAST_{<u}$,

¹⁴(34) is an example of a sentence that only makes sense on a simultaneous reading: the speaker’s past thought is said (by them) to be the reason why they were quiet. With some lexical and world knowledge, a learner will understand that the time of the thought and the time describing its content are the same.

¹⁵We must take it for granted that the learner notices the restricted licensing environment of $PAST_{\emptyset}$, but this is not because we think that this is in any way trivial.

then the two parts of (36) should have the back-shifted reading. This reading is available, but it is not the target reading either. The reading we want is simultaneous in the first half, and back-shifted in the second. Holding in place our assumptions about VP ellipsis, we do not know of a way to derive this reading without using $PAST_{<u}$. The LF is shown in (37).

- (37) LF: [my grandfather $PAST_{<u}$ think that [Nixon was $<u$ a good president]]]
[These days nobody $PRES_u$ think that [Nixon was $<u$ a good president]]]

Returning now to (35) and (36), we cannot rule out the possibility that both kinds of example occur in the learner's input, but we suspect that (36) is exceedingly rare. If we are right about this, then we see no reason for learners of English to posit $PAST_{<u}$ if they already have $PAST_{\emptyset}$; $PAST_{\emptyset}$ and $PAST_{<\emptyset}$ cover all the data we see above, except (36). For this reason, we are tempted to think that $PAST_{<u}$ is added to the lexicon—making $PAST_{<\emptyset/<u}$ —before $PAST_{\emptyset}$ is considered. As we suggested above, this may be because $PAST_{<u}$ is part of the initial state, and therefore the probability of using it remains relatively high—higher than $PAST_{\emptyset}$ —even after $PAST_{<\emptyset}$ becomes the favored lexical entry. $PAST_{\emptyset}$ is then added in response to examples that are incompatible with $PAST_{<\emptyset/<u}$, like (35).

From this, we derive a prediction. Since examples like (34) do not drive learners towards $PAST_{\emptyset}$, we expect to see languages where $PAST > PAST$ embeddings allow simultaneous readings, but which are not SOT languages. We do not know if such languages exist. There are scattered reports that some speakers of Russian permit the simultaneous readings of these embeddings in some cases, but we expect more robust judgements than that.

4.4 Russian

The Russian $PAST$, like Japanese, is uniformly $PAST_{<\emptyset}$ in the adult grammar, and the learner's path to it (away from the initial $PAST_{<u}$) is the same: embedded $PAST$ occurring under a forward-shifting temporal operator. The facts of Russian $PRES$ show another challenge, however. Recall that the present tense in Russian behaves like $PRES_{\emptyset}$ under attitude verbs and under other instances of $PRES$ (like the future marker), but is indexical ($PRES_u$) elsewhere. This is not true of English or of Japanese. Since neither $PRES_{\emptyset}$ nor $PRES_u$ is compatible with all of the observable input—the input to the Russian learner, that is—we speculate that the learner eventually adopts multiple entries for the present: $PRES_{\emptyset/u}$ with disjoint licensing conditions.¹⁶ Crucially, however, success in reaching this state depends on what hypotheses are available and what hypotheses are not. In other work on Russian tense the present tense is characterized as being “perspectival,” that is, as denoting the time of the local perspective. Under attitude verbs, the local perspective is that of the attitude holder, so $PRES$ comes to denote $\emptyset$. In other environments, e.g. at matrix level and inside RCs, the local perspective is that of the speaker, so $PRES$ comes to denote u . Call this lexical entry of the present tense $PRES_p$. From the perspective of semantic theory, $PRES_p$ is more interesting than the ambiguity-based account we have considered here. For one

¹⁶Again, we have nothing to say about how the learner acquires these licensing conditions.

thing, $PRES_p$ has precedent in other perspective-sensitive elements in natural language.¹⁷ Moreover, a single element $PRES_p$ (rather than the homophonous $PRES_{\emptyset/u}$) captures the distribution of licensing the different readings; the learner does not need to learn it separately/conditionally. Nevertheless, we see a potential problem for acquisition in $PRES_p$. If the hypothesis is attempted by a learner of English, no positive input can move them away from it. As we explained in the previous section, the double-access reading (and the situations that call for the use of double-access sentences in English) are compatible with a $\emptyset$ -interpretation of the embedded present. A learner that has $PRES_p$ therefore has no reason to change their semantics in response to such sentence-situation pairs. But we know that the English present tense is *not* perspectival (recall the contrast between (18) and (19)). Since (we assume) all English learners acquire this detail, we conclude that $PRES_p$ was never entertained by them. This may be because $PRES_p$ is not in the hypothesis space at all (and so not the right analysis of Russian), but it also may be that $PRES_p$ is special in some way, for example as a response only to the same kind of instability that leads our learner to posit (conditioned) ambiguity. For the Russian learner, the instability is the need to switch between $PRES_{\emptyset}$ (in response to attitude verb embeddings) and $PRES_u$ (in response to RC embeddings). English learners do not experience this, since both embeddings are compatible with $PRES_u$. If something along these lines is on the right track, the takeaway is that $PRES_p$ is different from $PRES_u$ or $PRES_{\emptyset}$ in its accessibility to learners. Testing this conclusion with Russian-learning children, say by looking at their command of PRES at different stages, is something that we leave to future work.

5. Discussion

We have proposed a framework¹⁸ for the acquisition of tense semantics which proceeds by incremental, local updates to scored form-meaning entries, with optionality and conditional licensing handled via lexical ambiguity. Applying this framework, we have identified (if our assumptions are correct) a path from an initial state — $PAST_{<u}$ - $PRES_{\emptyset}$ — to the adult grammars of Japanese, English, and Russian, locating the differences among the three trajectories in the input that each language supplies.

By considering semantic theory and acquisition in tandem, new insights arise. While intuition suggests that the availability of simultaneous readings of $PAST > PAST$ sentences would play a critical role in learning SOT — where the rule most clearly applies — such sentences cannot provide evidence in favor of SOT from the learner’s initial state of $PAST_{<u}$. The $\emptyset$ -past can be motivated only after the indexical past has been displaced, and the inputs that may displace it — embeddings of PAST under future operators — are of an entirely different kind. The logic of our account imposes an ordered sequence on acquisition and identifies critical inputs that are not made obvious by inspection of the adult grammar.

Which of these critical inputs occurs with reasonable regularity in the input is now an empirical question that must be addressed. Should a subset thereof prove vanishingly rare, the account cannot stand as given. Either one of the assumptions will need to be

¹⁷An example is the first person pronoun in Uyghur. See Shklovsky and Sudo 2014 for discussion.

¹⁸Doubtless a more rigorous treatment than space allows for here would require implementing the framework computationally to formally test our prose assertions via simulation.

amended (perhaps the initial state is different from what Hollebrandse's and Demirdache and Lungu's findings would suggest) or perhaps children do not learn the semantics of embedded tense *directly*, but acquire it instead as a by-product of some change prompted by another construction.

Leaving such issues aside for the moment, a consequence of our commitment to local updating is that learners cannot exploit *indirect* negative evidence (the inference that a form or interpretation which fails to occur, despite opportunity, is thereby disallowed).¹⁹ Detecting such an absence requires either surveying a stored record of past input for a distributional gap, or maintaining an expectation and registering its repeated non-confirmation. The former possibility is ruled out by incrementality as there is no retained distribution of form-meaning pairs in which a gap could be found. The proposed updating procedure prevents the latter possibility. The score for a hypothesis, $c(r_s)$, only changes in response to an observed input that the hypothesized entry does or does not license, never in response to non-occurrence. We emphasize that the framework does not stipulate a prohibition on indirect negative evidence. We motivated independent assumptions about memory and evaluation, and the positive-evidence-only character of the learner emerges from them.

A reader may notice that this creates an equivalence class of possible grammars. The learning procedure aims for convergence on a sufficiently supported grammar, not on a unique one, and so which grammar the learner ultimately decides on depends on which hypotheses are entertained in light of the input. Here a fact about the population is telling. Where the input along with the learning procedure genuinely underdetermine a grammar, learners are known to diverge, distributing across the compatible options (Han et al. 2007). In the case of the semantics of tense we know of no such variation. This suggests that the space of possible grammars is strongly constrained by some general principles (as yet unknown) for how entries are generated and put in competition.

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¹⁹For a broader discussion of indirect negative evidence within acquisition see Yang (2017).

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